SPREADSHEET FUNDAMENTALS

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In this section we'll cover **spreadsheet fundamentals**, including data literacy concepts like tables, cell references, data types, duplicate values, sorting & filtering, and more

TOPICS WE'LL COVER:

- Data Terminology
- Data Types
- Data Validation
- Removing Duplicates
- Conditional Formatting
- Sorting & Filtering

GOALS FOR THIS SECTION:

- Learn to navigate across cells in a spreadsheet and move, create, and delete rows & columns
- Enter, edit, and format cell values with text, numbers, dates, dropdowns, and checkboxes
- Remove duplicate values in one or more columns
- Apply conditional formatting rules
- Sort & filter data in a table

THE SECTION PROJECT

THE SITUATION

You own and operate **Flamin' Maven**, a food truck in Phoenix that serves authentic Mexican food from Thursday to Sunday each week

THE BRIEF

You use Excel to keep a digital record of the **restaurant checks**, including the date, party size, bill amount, tip, payment type, and server for each order

Your goal is to **leverage your Excel skills** to organize and explore the data in order to facilitate data entry, find interesting patterns, and make informed decisions

THE OBJECTIVE

Use Excel to:

- Make the data easier to read by applying formatting
- Facilitate data entry by using data validation rules
- Verify service and earnings by applying custom filters

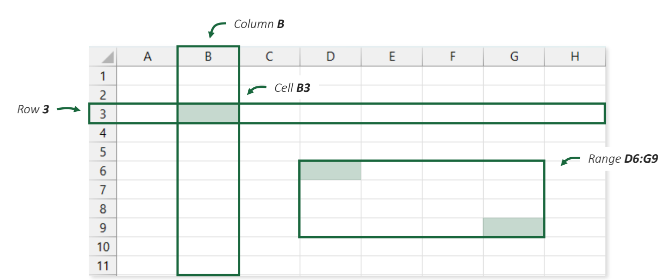
FLAMIN MAVEN

CELLS, RANGES, ROWS & COLUMNS

- **Data Terminology**
- Data Types
- Data Validation
- Removing Duplicates
- Conditional Formatting
- Sorting & Filtering

A spreadsheet is a grid of **cells** arranged into **rows** & **columns**

- **Rows** are represented by numbers *(1, 2, etc.)*, and **columns** are represented by letters *(A, B, etc.)*
- A **cell** is represented by the column letter and row number in which it lies *(A1, C4, etc.)*
- A **range** is a continuous group of cells represented by the top left and bottom right cells *(A1:C4)*



COMMON DATA TERMINOLOGY

When **data** is organized into the rows & columns of a spreadsheet, there are several terms used to describe its elements:

*A column that contains the same type of data is called a **field**, variable or feature*

*A row that contains the attributes for a single entity is called a **record** or observation*

	A	B	C	D	E	F	G	H	I
1	Check #	Date	Day	Time	Party Size	Total Bill	Tip %	Cash	Server
2	1	1/2/2025	Thu	Lunch	2	19.3	0.2	FALSE	Dennis
3	2	1/2/2025	Thu	Lunch	3	29.51	0.2	TRUE	Charlie
4	3	1/2/2025	Thu	Lunch	3	37.95	0.18	FALSE	Dennis
5	4	1/2/2025	Thu	Lunch	2	33.91	0.2	FALSE	Mac
6	5	1/2/2025	Thu	Lunch	4	69.37	0.2	FALSE	Dennis
7	6	1/2/2025	Thu	Lunch	4	52.8	0.15	TRUE	Charlie
8	7	1/2/2025	Thu	Lunch	2	26.56	0.2	TRUE	Charlie
9	8	1/2/2025	Thu	Lunch	4	28.17	0.18	FALSE	Dennis
10	9	1/2/2025	Thu	Lunch	2	27.36	0.18	FALSE	Dee
11	10	1/2/2025	Thu	Lunch	2	32.07	0.18	FALSE	Mac
12	11	1/2/2025	Thu	Lunch	2	27.14	0.2	TRUE	Dee

*A collection of related fields and records is a **table***

DATA TYPES

A cell can only store data of a single **data type**, which can be one of the following:

- **Numeric**: whole or decimal numbers that can be used in calculations *(adding, multiplying, etc.)*
- **Date**: calendar date & times that can be broken down into components *(year, month, hour, etc.)*
- **Binary**: TRUE or FALSE values *(can also be interpreted as 1 or 0)*
- **Text**: any other combination of letters and characters

*Numbers & dates are **right-aligned** by default*

*Text is **left-aligned***

*Binary values are **centered***

Check #	Date	Day	Time	Server	Party Size	Total Bill	Tip %	Cash
1	1/2/2025	Thu	Lunch	Dennis	2	19.3	0.2	FALSE
2	1/2/2025	Thu	Lunch	Charlie	3	29.51	0.2	TRUE
3	1/2/2025	Thu	Lunch	Dennis	3	37.95	0.18	FALSE
4	1/2/2025	Thu	Lunch	Mac	2	33.91	0.2	FALSE
5	1/2/2025	Thu	Lunch	Dennis	4	69.37	0.2	FALSE
6	1/2/2025	Thu	Lunch	Charlie	4	52.8	0.15	TRUE
7	1/2/2025	Thu	Lunch	Charlie	2	26.56	0.2	TRUE
8	1/2/2025	Thu	Lunch	Dennis	4	28.17	0.18	FALSE

DATE VALUES

Every date in Excel has an associated numerical **date value**

- Excel treats **January 1, 1900** as day 1, then counts each day that passes after that
- Times are captured as decimal values *(fractions of a 24-hour day)*, starting at midnight

This is what you see

Date
1900-01-01
1900-01-02
2021-05-29
2023-04-22 12:00:00
2023-04-22 18:00:00

This is what Excel sees

Date
1
2
44345
45038.50
45038.75

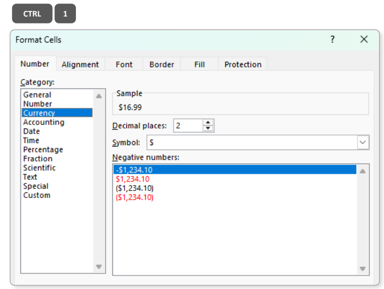
HEY THIS IS IMPORTANT!

Excel does not recognize dates before 1900, and will treat them as text

NUMBER FORMATTING

Excel contains a variety of predefined **number formatting** options:

CTRL 1



***Number formatting** lets you:*

- *Add a thousands separator and currency symbols*
- *Round to whole numbers or specific decimal places*
- *Show numbers as percentages*
- *Display dates & times in different formats*

HEY THIS IS IMPORTANT!

The underlying value stays the same, you're only changing the way it's shown